

Student Record Management System (Python + SQLite)

1. System Overview

CLI-based application to manage student records using SQLite. Supports full CRUD + search operations.

2. Database Schema

Table: students

```
CREATE TABLE students (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL,  
    age INTEGER,  
    course TEXT,  
    marks REAL  
);
```

3. Core Operations

Feature	SQL Operation
Add Student	INSERT
View Students	SELECT
Update Student	UPDATE
Delete Student	DELETE
Search Student	SELECT (WHERE LIKE)

4. System Flow

```
Start  
↓  
Connect DB  
↓  
Create Table  
↓  
Loop:  
    Show Menu  
    Take Input  
    Call Function  
    Execute SQL  
    Commit  
↓  
Exit → Close DB
```

5. Functional Modules

Database Layer: connect_db(), create_table()

Student Operations: `add_student()`, `view_students()`, `update_student()`, `delete_student()`, `search_student()`

Control Layer: `main()`, `close_connection()`

6. Input Requirements

name (string), age (integer), course (string), marks (float)

7. Output Requirements

Display records, confirmation messages, error handling

8. Constraints & Validations

Valid ID, age > 0, marks 0–100, name not empty

9. Edge Cases

Empty DB, invalid ID, duplicate names, no search results

10. Tech Stack

Python (core), SQLite (sqlite3 module)

11. Deliverable

Single .py file, auto-created DB, CLI interaction

12. Objective

Demonstrate Python + SQL integration, structured flow, data persistence